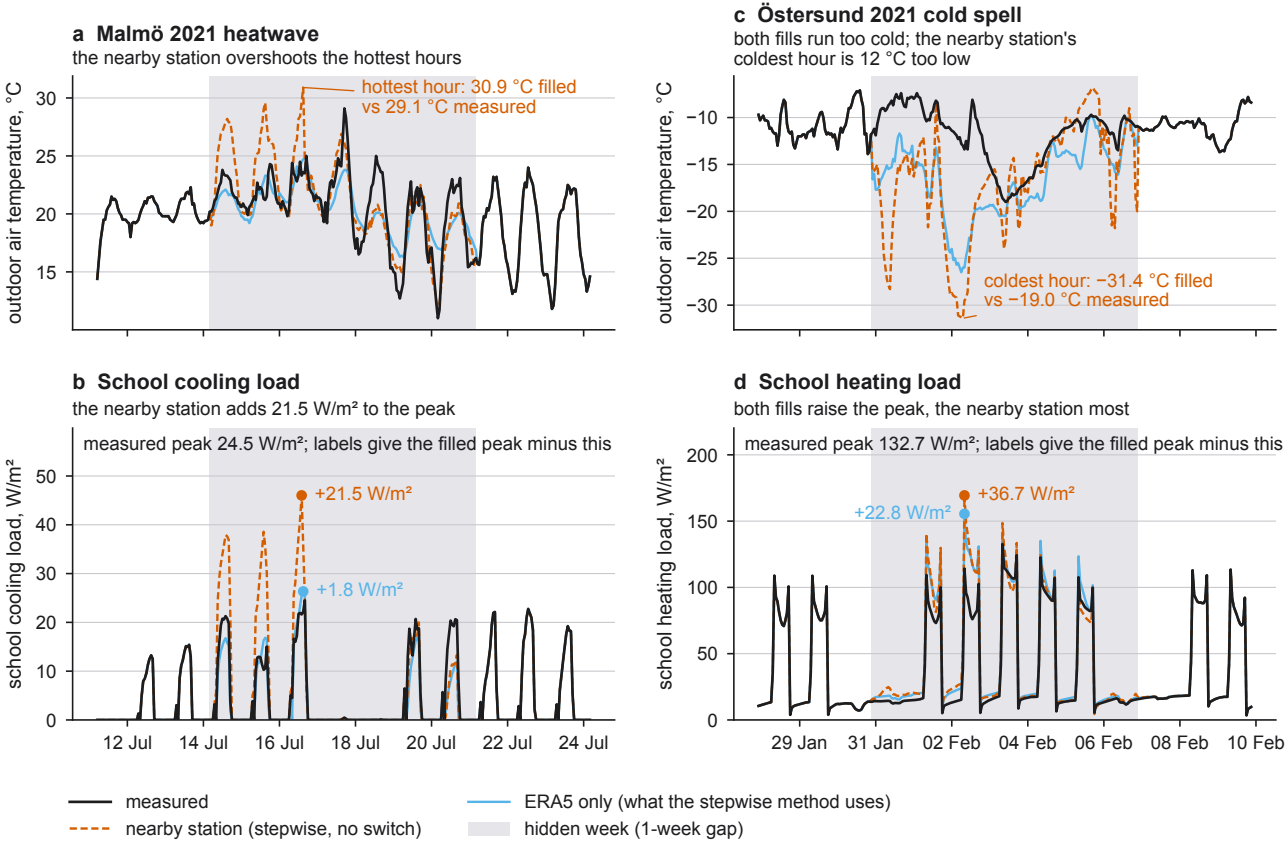

a Malmö 2021 heatwave
the nearby station overshoots the hottest hours
hottest hour: 30.9 °C filled
vs 29.1 °C measured
outdoor air temperature, °C
30
25
20
15
c Östersund 2021 cold spell
both fills run too cold; the nearby station's
coldest hour is 12 °C too low
coldest hour: −31.4 °C filled
vs −19.0 °C measured
outdoor air temperature, °C
−10
−15
−20
−25
−30
b School cooling load
the nearby station adds 21.5 W/m² to the peak
measured peak 24.5 W/m²; labels give the filled peak minus this
+21.5 W/m²
+1.8 W/m²
school cooling load, W/m²
50
40
30
20
10
0
12 Jul
14 Jul
16 Jul
18 Jul
20 Jul
22 Jul
24 Jul
d School heating load
both fills raise the peak, the nearby station most
measured peak 132.7 W/m²; labels give the filled peak minus this
+36.7 W/m²
+22.8 W/m²
school heating load, W/m²
200
150
100
50
0
29 Jan
31 Jan
02 Feb
04 Feb
06 Feb
08 Feb
10 Feb
measured
nearby station (stepwise, no switch)
ERA5 only (what the stepwise method uses)
hidden week (1-week gap)